Name - Naitik Dhima
Roll_no - 33
Section - Cyber.

Assignment - 2

Course: Foundation of Cyber Security (TCS 495)

Ques-1 Explain the importance of computer networking in ethical Hacking - Why must a cybersecurity student understand TCP/IP? Give five reasons:

Ans Computer networking is fundamental to ethical hacking as it enables the under of how data is transmitted, how systems communicate and how vulnerabilities can be exploited. A Cyber Security student must understand TCP/IP for the following reasons:

- TCP/IP is the foundational protocol suite for the Internet, making it essential for network communication.
- Understanding TCP/IP helps in identifying and mitigating network vulnerabilities
- Knowledge TCP/IP aids in analyzing network traffic and detecting anomalies.
- It is crucial for configuring firewalls and intrusion detection systems.
- Proficiency in TCP/IP is necessary for effective penetration testing and Vulnerability assessments.

Ques-2 Differentiate between the following virtual machine network modes in tabular form: Nat, Bridged, Host-only.

Network Mode	Description	Use Case.
NAT	Network Address Translation allows Vm's to share the host's IP address.	Accessing the internet from a VM without exposing it directly
Bridged	Connects the VM. Directly to the physical Network allowing it to have its Own IP address.	Integrating a VM into an existing network for testing.
Host-only	Creates a network that is accessible only between the host and the VM.	Isolated testing environment without external network access.

Ques-3 Explain the TCP-3 Way Handshake with diagram and Packet flow. What risks arise if handshake security is weak?

Ans

(ACK) —— (Client) —— (SYN)

The TCP-3 Way handshake consist of three steps:

1.) The Client sends a SYN packet to the server to initiate a connection.

2.) The server ~~sends~~ responds with a SYN-ACK Packet to acknowledge the request.

3.) The Client sends an ACK Packet back to the server to establish the connection.

Risks of weak handshake security include:

- Man in the middle Attacks where attacker intercepts the handshake.
- Session hijacking where an attacker takes over an established session.
- Denial of Service (DOS) attacks by overwhelming the server with SYN requests.

Ques-4 Explain the difference between: IPv4 VS IPv6 , IP Address v MAC Address DNS

Concept	IPv4	IP v.6
Address length	32 bits	128 bits
Address Format	Decimal (eg., 192.168.1.1)	Hexadecimal (eg, 2001:0db8:85a3:0000:0000:8a2e:0370:7334)
Address Space	Approximately 4.3 billion address.	Approximately 340 undecillion addresses.
Header Complexity	less complexity	More complexity with additional features
Usage	Still widely used.	Gradually replacing IPv4

Name - Nathk
Roll no -
Section

IP Address vs MAC Address

- IP Address: logical address assigned to a device on a network, can change based on the network.
- MAC Address: Physical Address hardcoded into the network interface and (NIC) unique to each device.

DNS vs Ping

- DNS (Domain Name System): Translates domain names into IP addresses.
- Ping: A network utility that tests the reachability of a host on an IP network.

Ques-5 A company is assigned IP address 192.168.10.0/24 and needs 4 departments with equal hosts. Find:

a) New subnet mask.
b) Number of subnets created.
c) Number of usable hosts per subnet.
d) First 4 subnet addresses

Ans To create 4 equal subnets from a /24 network:

- New subnet mask: /26 (since 2 bits are borrowed for subnetting).
- Number of subnets created: 4 ($2^2 = 4$).
- Number of usable hosts per subnet: 62 ($2^{(32-26)} - 2 = 62$).
- First 4 subnet address:

1. 192.168.10.0/26
2. 192.168.10.64/26
3. 192.168.10.128/26
4. 192.168.10.192/26

Ques 6 Given IP address 172.16.25.130/26, determine:

a) Network address.
b) Broadcast address.
c) Usable host range.
d) Total usable hosts.

Ans For the IP address 172.16.25.130/26:

- Network address : 172.16.25.128
- Broadcast address : 172.16.25.191
- Usable host range : 172.16.25.129 to 172.16.25.190
- Total usable hosts : 62 (2^(32-26) - 2 = 62)

Ques 7 Install any virtualization software such as Oracle VM Virtual Box or VMware Workstation and create one Linux VM. Write steps with screenshots for:
→ Installation, VM creation, RAM and storage allocation,
· Changing network adapter to NAT, Changing network to Bridged

Ans steps for installation and VM creation:

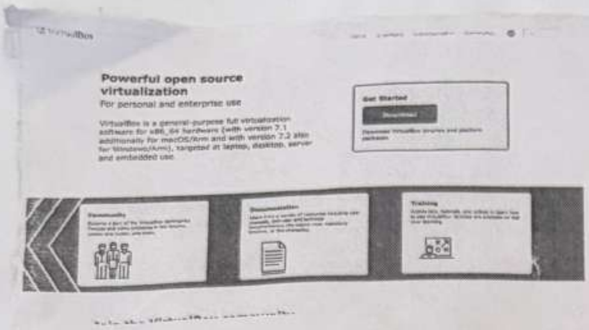
1) download and install Oracle VM Virtual Box from the official website

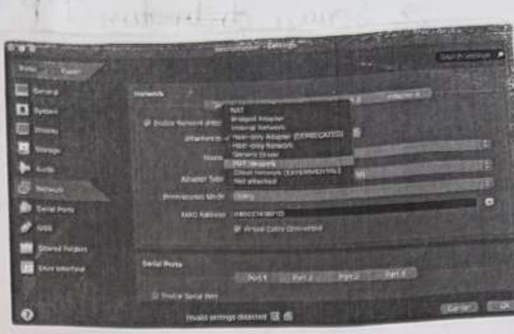
2) Open Virtual Box and click a New to create a new VM.

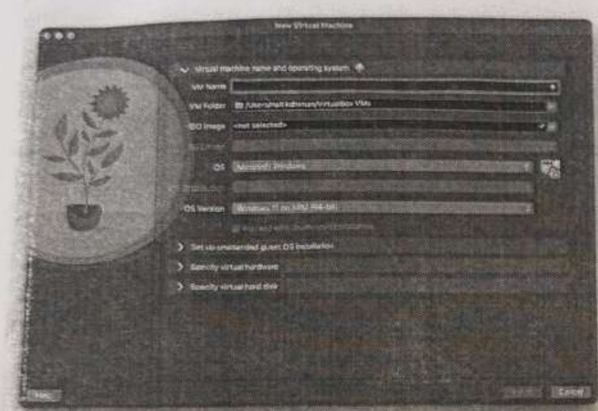
3) Allocate RAM (eg. 2048 MB) and storage (eg 20 GB).

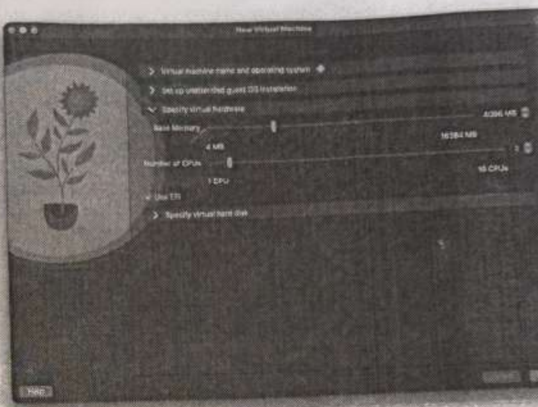
4) Change the network adapter to NAT in the settings.

5) Change the network adapter to Bridged in the settings.









Ques-8 Explore Wireshark and perform packet capture on your machine Capture DNS or Ping traffic and answer.

a) Source IP b) Destination IP c) Protocol used

d) Packet count.

Ans Wireshark is a network protocol analyzer that captures and displays packet data real time. To capture DNS or Ping traffic-

1) Open Wireshark and select the appropriate network interface (eth 0)
2) start the capture and perform a DNS query or Ping.
3) Stop the capture after sufficient data is collected,

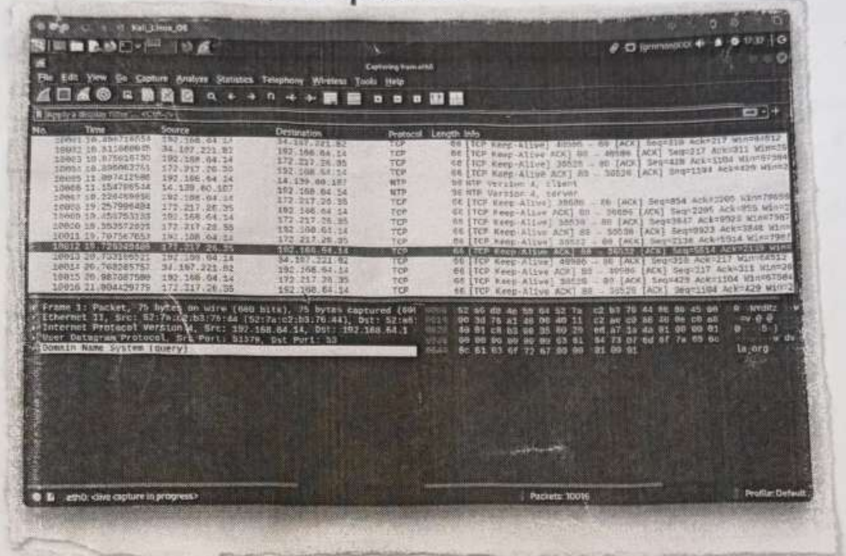
Results:

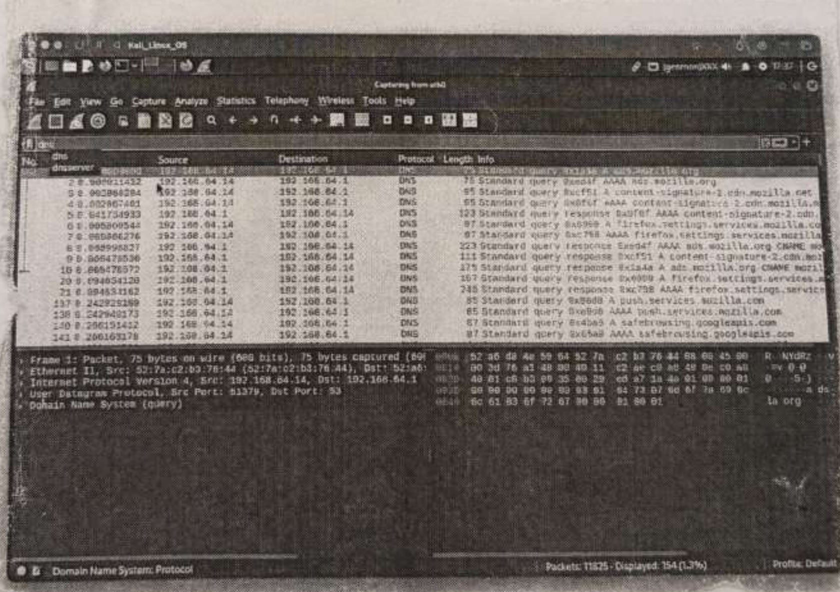
→ Source ~~destination~~ IP : 192.168.64.1 → 142.250.182.238 (google)

→ Destination IP : 192.168.64.3

→ Protocol used : DNS

→ Packet count : ~1100 packets.





Ques 9. Create a HTML page and Java Script program for a student registration form with input validation for:

- Name cannot be empty.
- Email format valid.
- Password minimum 8 characters.
- Mobile must be 10 digits.

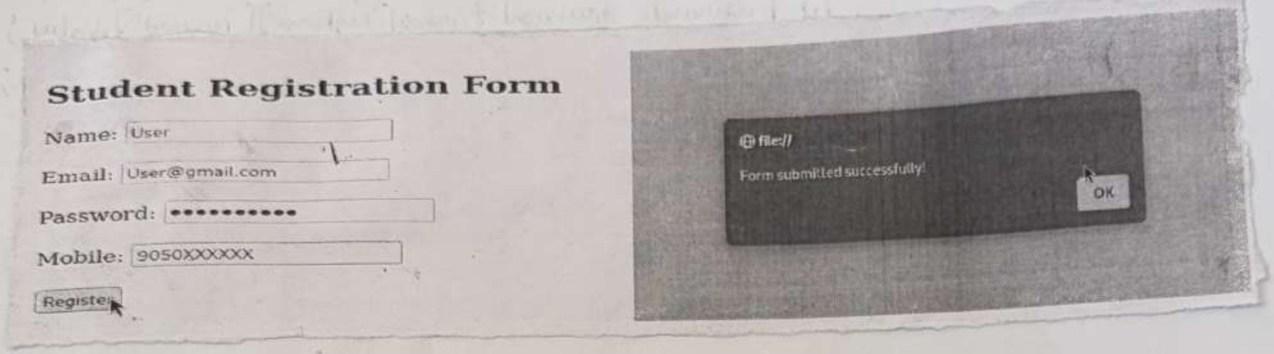
Also explain how poor validation can lead to XSS attacks.

```
<!DOCTYPE html>
<html>
<head>
    <title> Student Registration </title>
    <script>
        function validateForm() {
            let name = document.forms["regForm"]["name"].value;
            let email = document.forms["regForm"]["email"].value;
            let password = document.forms["regForm"]["password"].value;
            let mobile = document.forms["regForm"]["mobile"].value;
            let emailPattern = /^[^ ]+@[^ ]+\.[a-z]{2,3}$/;

            if (name == "") {
                alert ("Name cannot be empty");
                return false;
            }

            if (!email.match (emailPattern)) {
                alert ("Invalid Email Format");
                return false;
            }
            if ( password.length < 8) {
                alert ("Password must be at least 8 characters");
                return false;
            }
            if ( !/^\d{10}$/.test (mobile)) {
                alert (" Mobile must be 10 digits");
                return false;
            }
            alert ("Form submitted successfully !");
            return true;
        }
    </script>
</head>
<body>
    <h2> Student Registration Form </h2>
    <form name = "regForm" onsubmit = "return validateForm()">
        Name : <input type = "text" name = "name" > <br><br>
        Email : <input type = "text" name = "email" > <br><br>
        Password : < input type = "password" name = "password" > <br><br>
        Mobile : < input type = "text" name = "mobile" > <br><br>
        <input type = "submit" Value = "Register" >
    </form>
</body>
</Html>
```

Poor validation can lead to XSS (Cross-Site Scripting) attacks where attackers inject malicious scripts into web pages viewed by other users. If input is not properly sanitized, these scripts can execute in the control of the user's browser, leading to data theft or session hijacking.



Ques-10 Write a PHP program that connects to SQL database and perform secure login validation. Also answer:

a) What is SQL injection?

b) How prepared statements prevent it?

c) Difference between phishing and spear phishing.

Ans SQL Injection is a code injection technique that exploits vulnerabilities in an application software by inserting malicious SQL ~~injection~~ statements.

Prepared statement prevents SQL injection by separating SQL logic from data, ensuring that user input is treated as data only, not executable code.

Phishing is a cyber attack that attempts to trick individuals into providing sensitive information by masquerading as a trustworthy entity. Spear phishing targeted form of phishing where the attacker customizes their attack to a specific individual or organization increasing the likelihood of success.

```php
<?php
$db = new SQLite3('testdb.db');

$db->exec("Create Table if not Exists users (
        id Integer Primary Key Autoincrement,
        username Text,
        password Text
);");

$db->exec("Insert into users(username, password) Values
        ('admin', '1234')");

echo "Enter username: ";
$user = trim(fgets(STDIN));

echo "Enter password: ";
$pass = trim(fgets(STDIN));

$stmt = $db->prepare("Select * From users Where username = :user
        And password = :pass");
$stmt->bindValue(':user', $user, SQLITE3_TEXT);
$stmt->bindValue(':pass', $pass, SQLITE3_TEXT);
$result = $stmt->execute();

$row = $result->fetchArray();

if ($row) {
    echo "Login Successful\n";
} else {
    echo "Invalid Credentials\n";
}
?>
```

```
naitikdhiman@naitiks-Laptop Cyber_Notes % php run.php
Enter username: user
Enter password: 1234
Invalid Credentials
naitikdhiman@naitiks-Laptop Cyber_Notes % php run.php
Enter username: admin
Enter password: 1234
Login Successful
naitikdhiman@naitiks-Laptop Cyber_Notes %
```

Bonus Challenge

Create a mini Project titled Cyber Network Detector using any language. that perform any three tasks.

Ping scanner.
Port checker.
DNS look up.

Ans Code -

```
import os
import socket.
def. Ping(host):
    print("\n [+] Pinging target. ...")
    response = os.system(f"Ping -c 1 {host}")
    if {response == 0;
        print(f"{host} is Active")
    else:
        print(f"{host} is InActive")

def port_check (host, port):
    print ("\n [+] Checking port. ..").
    s = socket.socket()
    s.settimeout (2)
    result = s.connect_ex((host, port))
    if result == 0:
        print(f"Port {port} is OPEN on {host} ")
    else:
        print(f"Port {port} is CLOSED on {host}")
    s.close()
```

```
def dns_laakup (domain):
    print ("\n [+] Perfoming DNS Lookup ---")
    try:
        ip = socket . gethostbyname(domain)
        print (f"IP Address of {domain} is {ip} ")
    except:
        print. ("Invalid domain or unable to resolve")

while True:
    Print ("\n ===== Cyber Network Detector ====")
    Print ("1. Ping Scanner")
    print ("2. Port Checker")
    print ("3. DNS Lookup")
    Print ("4. Exit")
    choice = input ("Enter your choice : ")
    if choice == '1':
        target = input("Enter ~~IP or~~ Domain:")
        Ping (targe
    elif choice =='2':
        target = input ("Enter IP or Domain: ")
        port = int (input ("Enter Port Number : "))
        Port checker ( target, Port )

    elif choice =='3':
        domain = input ("Enter Domain Name : ")
        dns_lookup (domain)
    elif choice == '4':
        print ("Exiting tool...")
        break.
    else:
        print ("Invalid choice, try again.")
```

```
naitikdhiman — -zsh — 80×24

Last login: Tue May  5 17:07:52 on ttys003
naitikdhiman@naitiks-Laptop ~ % ping google.com
PING google.com (142.250.182.46): 56 data bytes
64 bytes from 142.250.182.46: icmp_seq=0 ttl=119 time=17.054 ms
64 bytes from 142.250.182.46: icmp_seq=1 ttl=119 time=24.553 ms
64 bytes from 142.250.182.46: icmp_seq=2 ttl=119 time=26.918 ms
64 bytes from 142.250.182.46: icmp_seq=3 ttl=119 time=20.749 ms
64 bytes from 142.250.182.46: icmp_seq=4 ttl=119 time=31.849 ms
64 bytes from 142.250.182.46: icmp_seq=5 ttl=119 time=33.233 ms
^C
--- google.com ping statistics ---
6 packets transmitted, 6 packets received, 0.0% packet loss
round-trip min/avg/max/stddev = 17.054/25.726/33.233/5.724 ms
naitikdhiman@naitiks-Laptop ~ %
```

```
┌──(kali㉿kali)-[~/Assignment1]
└─$ python3 scrip6.py

===== Cyber Network Detective =====
1. Ping Scanner
2. Port Checker
3. DNS Lookup
4. Exit
Enter your choice: 1
Enter IP or Domain: google.com

[+] Pinging target ...
PING google.com (142.250.182.238) 56(84) bytes of data.
64 bytes from tzdelb-bd-in-f14.1e100.net (142.250.182.238): icmp_seq=1 ttl=118 time=18.7 ms

--- google.com ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 18.683/18.683/18.683/0.000 ms
google.com is Active

===== Cyber Network Detective =====
1. Ping Scanner
2. Port Checker
3. DNS Lookup
4. Exit
Enter your choice: 2
Enter IP or Domain: google.com
Enter Port Number: 22

[+] Checking port ...
Port 22 is CLOSED on google.com

===== Cyber Network Detective =====
1. Ping Scanner
2. Port Checker
3. DNS Lookup
4. Exit
Enter your choice: 3
Enter Domain Name: google.com

[+] Performing DNS Lookup ...
IP Address of google.com is 142.250.182.238
```